

EXP 24 – Diet Recommendation System (Prolog)

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Aim

To write a Prolog program that suggests an appropriate diet plan based on a given disease by storing diet recommendations as facts and retrieving them using rules.

Algorithm

1. Define diet recommendations for each disease using diet(Disease, ListOfSuggestions).
2. When the user queries a disease, retrieve the corresponding list of suggestions.
3. Display each suggestion using format and forall.
4. If the disease exists in the knowledge base, print all diet guidelines for that disease.

Source Code

```
prolog
```

```
diet(diabetes, [
```

```
    'Avoid sugar and sweets',
```

```
    'Eat more leafy vegetables',
```

```
    'Prefer whole grains over refined carbs',
```

```
    'Avoid sugary fruits'
```

```
]).
```

```
diet(hypertension, [
```

```
    'Reduce salt intake',
```

```
    'Eat more potassium-rich foods',
```

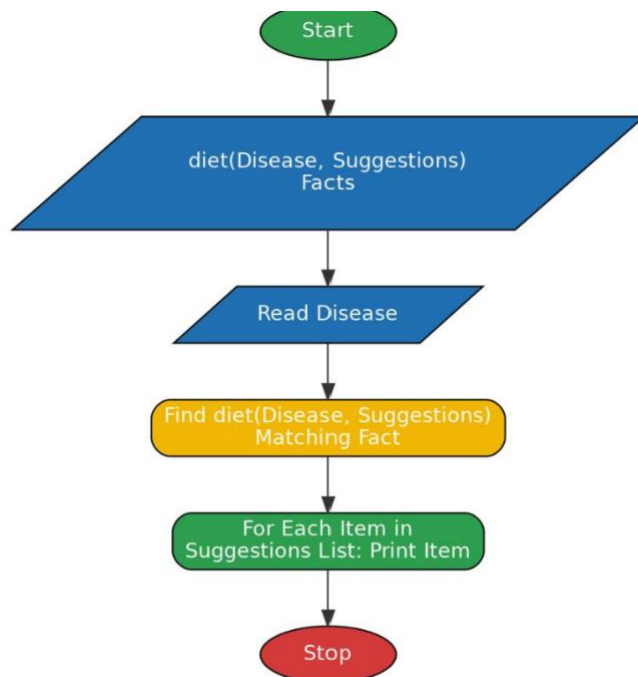
```
    'Avoid processed and canned food',
```

```
'Limit caffeine and alcohol'  
]).
```

```
diet(obesity, [  
    'Follow a low-calorie diet',  
    'Avoid fried and fatty foods',  
    'Increase fiber intake',  
    'Exercise along with a balanced diet'  
]).
```

```
suggest_diet(Disease) :-  
    diet(Disease, Suggestions),  
    format('Diet suggestions for ~w:~n', [Disease]),  
    forall(member(S, Suggestions),  
        format('- ~w~n', [S])).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [diet].
true.

?- suggest_diet(diabetes).
Diet suggestions for diabetes:
- Avoid sugar and sweets
- Eat more leafy vegetables
- Prefer whole grains over refined carbs
- Avoid sugary fruits
true.

?- suggest_diet(hypertension).
Diet suggestions for hypertension:
- Reduce salt intake
- Eat more potassium-rich foods
- Avoid processed and canned food
- Limit caffeine and alcohol
true.

?- suggest_diet(obesity).
Diet suggestions for obesity:
- Follow a low-calorie diet
- Avoid fried and fatty foods
- Increase fiber intake
- Exercise along with a balanced diet
true.
```

Result

The program successfully retrieves and displays diet recommendations for any disease defined in the knowledge base. When a disease is queried, it prints a clear list of dietary guidelines suitable for that condition.